

A Truthful Combinatorial Reverse Auction Mechanism for Crowdshipping

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Abstract—Crowdshipping is an emerging logistics paradigm that leverages the excess capacities of occasional couriers (OC) to transport goods. This article investigates the problem that a crowdshipping platform recruits OCs to serve the parcel pickup and delivery requests. A combinatorial reverse auction framework is introduced to model the interactions between the platform and couriers. We design an auction mechanism that implements in polynomial time, which satisfies the computational requirements of crowdshipping. It can also achieve near-optimal social cost with the approximation ratio rigidly proved. Through both rigorous theoretical analyses and extensive simulations, we demonstrate that the proposed auction mechanism is truthful, individual rational, and has high computation efficiency.

Index Terms—Combinatorial reverse auction, crowdshipping, truthful mechanism.

I. INTRODUCTION

CROWDSIPPING, also known as *crowdsourced delivery*, is an emerging application of the concept “sharing economy” on the transportation sector, and has attracted increasing attention in academia and industry [1]. It refers to the participation of the crowd as occasional couriers (OCs) in transporting parcels along their journeys that already take place. Different from traditional transportation services fulfilled by creating new traffic flows, it exploits existing ones. Therefore, crowdshipping has been regarded as a new logistic solution that is both environmentally and socially friendly [2].

Despite that crowdshipping is a novel paradigm of logistics, its compensation schemes still remain old-fashioned: OCs are usually paid a fixed amount per delivery or per hour

(e.g., Walmart adopts per delivery payment and Amazon Flex employs hourly rates) [3]. However, couriers may use different modes of transportation, such as cars, trucks, scooters, or even public transit [4] to perform the delivery. Besides, for each parcel, the extra traveling distance due to detour varies for different couriers, while parcels themselves differ in their value and difficulty of delivery. Therefore, the costs of performing crowdshipping are highly diverse for different couriers and different parcels. Fixed payment schemes, although simple, fail to reflect such needs. Due to the large number of potential couriers, it is practically impossible for the crowdshipping platform to design a personalized payment scheme for an individual courier. A proper way is to allow couriers to submit bids which consist of the delivery requests they prefer to serve and the corresponding costs, and then the platform decides the winners and payments according to certain auction rules.

When designing auction mechanisms, several properties are expected to be satisfied: *truthfulness*—every bidder can obtain the largest utility by truthfully bidding their private information no matter how other bidders bid; *individual rationality*—the utility of any bidder in all possible outcomes cannot be smaller than those who do not enter the auction; and *social optimality*—minimizing the social cost which is the sum of the costs of OCs completing all the delivery requests. Individual rationality is necessary for motivating couriers to participate in the auction. The reason why truthfulness is desired is that couriers can achieve maximum utility simply by truthfully revealing their private information without considering the actions of others. Consequently, strategic behaviors no longer yield larger benefit, thus fairness and robustness of the auction are improved with the potential market manipulation kept away [5]. In addition, a social optimal auction that minimizes the social cost can enhance the stability and longevity of the crowdshipping platform [6], especially when facing fierce competition.

Different from traditional auction mechanisms applied in the transportation area, the novelty of crowdshipping introduces new challenges. In the environment of crowdshipping, a large number of couriers and requests arrive dynamically, thus an online scheduling with *fast computation speed* is required [7]. Moreover, the transportation cost can be subadditive [8], i.e., serving a package of requests can result in a lower total cost than the sum of costs of serving all requests individually. Therefore, *package bidding*, where each courier simultaneously bids for several preferred heterogeneous requests, will

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allow the couriers to fully express their preferences, and lead to improved economic welfare [9]. However, introducing package bidding yields a *combinatorial auction*, which is known for its high computational complexity [10]. Conventional mechanisms designed for combinatorial auction, such as the VCG mechanism, cannot be implemented in *polynomial time*, violating the computational requirement of crowdshipping. As a result, in this article, we pursue a combinatorial auction mechanism that can tackle with the challenges brought by the implementation of crowdshipping.

The main contributions in this article are summarized as follows.

- 1) We construct a novel computationally efficient combinatorial reverse auction framework¹ for the crowdshipping platform to recruit OCs in serving pickup and delivery requests.
- 2) Truthfulness, individual rationality, and polynomial-time computation complexity of the proposed mechanism are rigorously proved. The theoretical approximation ratio of the mechanism is also provided.

The remainder of this article is organized as follows. A review of related works is reported in Section II. Section III provides a detailed description of system models and problem formulation. In Section IV, the combinatorial reverse auction mechanism is designed, and theoretic analyses of its properties are performed. Numerical results are presented in Section V, followed by the conclusion in Section VI.

II. RELATED WORKS

This article is related to two main streams of research: 1) crowdshipping and 2) auction mechanisms implemented in the transportation area. Next, we compare this article with related works in detail.

A. Crowdshipping

The literature on crowdshipping is rapidly emerging in recent years. Archetti et al. [11] introduced the vehicle routing problem with occasional drivers (VRPODs), where a company uses both self-owned and occasional drivers (OD) to make deliveries. Torres et al. [12] extended VRPOD to consider stochastic destinations of ODs. Perboli et al. [13] further studied the dynamic and stochastic VRP with crowd drivers using a simulation-optimization approach. In [14], a rolling horizon approach is proposed to match the pickup and delivery tasks with crowdsourced drivers dynamically. Cheng et al. [15] presented an integer programming framework to exploit the underutilized capacity of crowdsourced public transportation systems to perform city-wide package delivery. Du et al. [16] proposed a spatial-crowdsourcing-based system to have packages take hitchhiking rides with existing taxis. In [17], an online dynamic optimization framework is developed to deliver on-demand food by crowdsourced public riders. In [18], private car trips are leveraged to incidentally deliver parcels during their original trips. Parcel delivery cost is estimated, and delivery tasks are optimally assigned to car

trips accordingly. Wang et al. [19] predicted the preference of crowdsourced riders and allocates orders to optimize the experience of the on-demand food delivery system. In [20], an economical approach to express package delivery is proposed by exploiting the crowd of taxis to help transport packages. Some works reviewed above neglect the financial compensation for crowdsourced couriers, while others simply use a fixed compensation scheme or a variable one proportional to the traveling distance of each delivery task.

In [21], four different compensation schemes are evaluated for the crowdshipping system to compare the profits of couriers and platform. In [22], a personal compensation scheme for crowdshippers is determined based on their acceptance behavior uncertainty, which is captured by the discrete choice model. Despite individual compensation scheme has been studied by these previous works, they ignore the potential strategic behavior of couriers. In other words, since the platform requires private information from couriers to determine the proper compensation scheme, couriers can report false information for their own benefit.

B. Auctions

Auctions have been widely applied in the transportation sector. For example, [6], [23], [24], [25], [26], and [27] consider the transportation service procurement problem using auction mechanisms, including combinatorial auctions. Zhang et al. [28], Lam [29], Karamanis et al. [30], and Bian et al. [31] studied the auction mechanism for ridesharing. In addition to their different application scenarios from crowdshipping, auction mechanisms in these works are different from the one proposed in this article. To be specific, [6], [23], [24], [27], and [29] all adopt the VCG-based mechanism that requires solving an NP-hard problem to optimality and thus is not suitable for the crowdshipping system with a large number of OCs.² In [25], an iterative auction mechanism is developed, but no guarantee of truthfulness or individual rationality is provided. Yu et al. [26] employed the multiattribute multiunit auction where the multiple units are homogeneous, while our proposed combinatorial auction mechanism considers multiple heterogeneous units. In [28], a double auction mechanism is designed to match passengers and drivers, nevertheless, bidders cannot submit bids for several preferred items simultaneously and thus is not a combinatorial auction. In [30], a combinatorial double auction is proposed, yet it is not truthful. In [31], the bidders are the passengers in first-mile ridesharing and the auction is therefore a forward auction. In comparison, bidders in our work are the couriers that sell services to the platform, rendering a reverse auction.

Moreover, the auction is a popular technique to handle challenges in other related areas as well. For example, [32] proposed a VCG combinatorial auction-based dynamic pricing scheme for the multigranularity service composition problem. As discussed above, the VCG mechanism is not suitable for the crowdshipping scenario due to computational constraints. In [33], an online combinatorial auction problem is studied,

¹For *reverse auctions*, the bidders are the sellers, while for *forward auctions*, bidders are the buyers.

²The detailed reasoning is presented by the end of Section III.

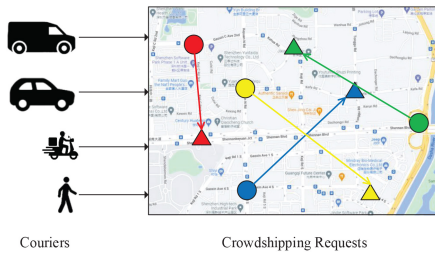


Fig. 1. Recruiting OCs to perform crowdshipping requests. The crowdshipping platform has a set of pickup and delivery requests that need to be served, where the circles represent pickup nodes and the triangles represent delivery nodes. It recruits OCs which are based on trucks, cars, scooters, and public transit to perform these requests.

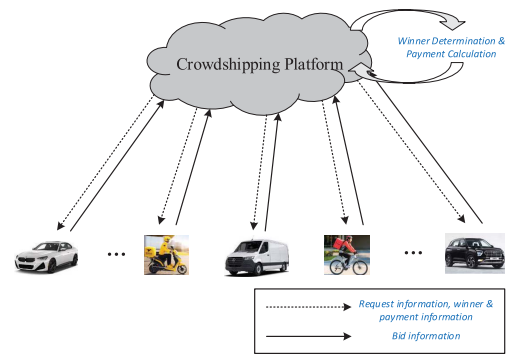


Fig. 2. Basic framework of the crowdshipping system.

and posted-price mechanisms are provided with optimal competitive ratios. However, [33] considers a forward auction with sequentially arriving customers, which is a different scenario from the problem we consider.

Among the previous literature, the most related ones with this work are the following three. In [34], a VCG-based multi-attribute multiunit auction is proposed for crowdsourced delivery, where the requester bids for courier services. As a result, it is also a forward auction and not applicable to our reverse auction setting. In [35], a multiattribute auction mechanism is designed for couriers in crowdsourced first and last mile logistics. However, parcels are auctioned one by one, hence it cannot capture the case where package bidding is preferred. In [36], occasional drivers are procured for freight transportation using a combinatorial auction. No performance guarantee like truthfulness and individual rationality is provided.

III. SYSTEM MODEL AND PROBLEM FORMULATION

We consider an online crowdshipping platform that recruits OCs to serve the parcel pickup and delivery requests, as shown in Fig. 1. The set of requests is denoted as $\mathcal{N} = \{1, 2, \dots, N\}$, and the set of available couriers is denoted as $\mathcal{K} = \{1, 2, \dots, K\}$. Viewing request information published by the crowdshipping platform, OCs select their preferred requests and submit bids to the platform. After that, based on the combinatorial reverse auction framework, the platform determines the winning bids and calculates the payments to the winning couriers. Detailed models of this crowdshipping system will be elaborated in the following sections.

A. Combinatorial Reverse Auction Framework

In the crowdshipping system shown in Fig. 2, couriers compete for opportunities to serve the requests. Suppose OCs have access to the request information through their mobile phone APPs. We introduce a *combinatorial reserve auction* (CRA) framework to model the interactions between the couriers and the crowdshipping platform. In this auction, couriers are *sellers*, the platform is the *buyer* that buys parcel delivery services from couriers.

As illustrated in Fig. 3, the CRA framework can be implemented as follows.

- 1) The crowdshipping platform publishes request information to the couriers through mobile phone

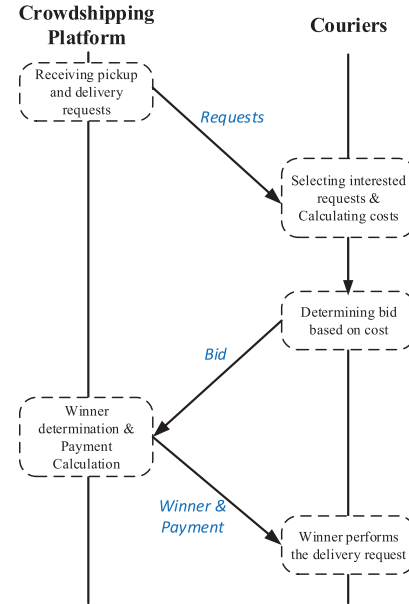


Fig. 3. Information flow between couriers and the platform within the combinatorial reverse auction framework.

APPs. The information includes pickup location and time, delivery location and time of each request, and so on.

- 2) Courier $k \in \mathcal{K}$ selects the preferred requests according to her own traveling origin and destination, and then calculates the traveling cost. Details for calculating costs will be specified in the next section. After that, she can determine the bid $\beta_k = \{Q_k, b_k\}$ to be submitted to the platform, where Q_k is the set of requests she prefers to serve, and b_k is the traveling cost she reports to the platform.
- 3) The platform determines whether a bid wins or not, i.e., it selects a winning subset S from the set of bids $B = \{\beta_1, \dots, \beta_K\}$, and then calculates payments for the winning bids.
- 4) Winning couriers perform the requests in their bids and receive payments from the platform accordingly.

In this auction framework, how to determine the winning bids and how to design payments are two key points to ensure the performance of the crowdshipping system. Assume that

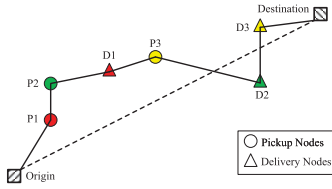


Fig. 4. Example of an OC.

each courier seeks to maximize her own benefit, which is formally defined as her utility:

Definition 1: The utility u_k of courier k is defined as the difference of received payment p_k and the traveling cost $\bar{b}_k$, i.e.,

$$u_k = p_k - \bar{b}_k. \quad (1)$$

Due to privacy issues, the exact traveling cost $\bar{b}_k$ of each courier k is unknown to others and the platform. Therefore, it is possible for strategic couriers to misreport the traveling cost b_k for her own good, i.e., the claimed cost b_k may not be the same with the true cost $\bar{b}_k$. In this way, the couriers can manipulate the obtained payments p_k to improve their utility. However, the existence of strategic behaviors endangers the equity and robustness of the system. As a result, we will pursue an auction mechanism that minimizes the total cost and encourages truth-telling and couriers' participation in the crowdshipping service.

B. Model of Occasional Couriers

As for the calculation of each courier's delivery cost, we need to introduce their mathematical models. Each courier $k \in \mathcal{K}$ has her own traveling origin and destination. The request information includes locations of pickup and delivery nodes, pickup and delivery time, the weight of the parcel to be delivered, and so on. After observing this information, a courier can select preferred requests, calculate the serving cost and then submit a bid to the platform.

For example, Fig. 4 depicts an OC. The dotted line is the previous traveling path from origin to destination. With the selected requests, a new traveling path is planned, represented by the solid line. That is, this courier starts from the origin, makes a detour to serve pickup and delivery pairs ($P1 - D1$), ($P2 - D2$), and ($P3 - D3$), and arrives at the destination in the end. The extra traveling cost due to crowdshipping is calculated as the difference of costs between the solid line and the dotted line. She then submits a bid to the platform based on the extra cost.

How to decide the optimal traveling path to serve crowdshipping requests and calculate the corresponding cost can be tackled by solving a pickup and delivery problem with time windows (PDPTWs) [37]. Before that, we first introduce some notations. Assume that courier k serves n_k requests. Then, we can denote the set of pickup nodes as $P_k = \{1, 2, \dots, n_k\}$, and the set of delivery nodes as $D_k = \{n_k + 1, \dots, 2n_k\}$. Further, let the set $N_k = P_k \cup D_k$, $V_k = N_k \cup \{0\} \cup \{2n_k + 1\}$, where $\{0\}$ and $\{2n_k + 1\}$ are the origin and destination, respectively.

TABLE I
PARAMETERS FOR THE PDPTW

d_{ij}	Distance between node i and j
t_{ij}	Traveling time between node i and j
s_i	Service time at node i
a_i	Opening time window at node i
b_i	Closing time window at node i
l_j	Increased load after visiting node j
$L_{\max}$	Loading capacity of courier
w_k^d	Weighting parameter for traveling distance of courier k
w_k^t	Weighting parameter for traveling time of courier k

For the variables of PDPTW, let x_{ij} be a binary variable which takes value 1 if the courier visits node j immediately after visiting node i , and takes value 0 otherwise. Assume the time courier visits node i is t_i , and the current load carried by the courier after visiting node i is L_i . Other parameters of PDPTW are listed in Table I. The PDPTW solved by each courier k is as follows:

$$\min_{x_{ij}, t_i, L_i} w_k^d \sum_{i,j} d_{ij} x_{ij} + w_k^t (t_{2n_k+1} - t_0) \quad (2a)$$

$$\text{s.t.} \sum_{j \in N_k} x_{ij} = 1 \quad \forall i \in P_k \quad (2b)$$

$$\sum_{j \in V_k} x_{ij} - \sum_{j \in V_k} x_{j,n+i} = 0 \quad \forall i \in P_k \quad (2c)$$

$$\sum_{j \in P_k \cup \{2n_k+1\}} x_{0j} = 1 \quad (2d)$$

$$\sum_{i \in D_k \cup \{0\}} x_{i,2n_k+1} = 1 \quad (2e)$$

$$\sum_{i \in V_k} x_{ij} - \sum_{i \in V_k} x_{ji} = 0 \quad \forall j \in N_k \quad (2f)$$

$$(t_i + s_i + t_{ij} - t_j) x_{ij} \leq 0 \quad \forall i, j \in V_k \quad (2g)$$

$$a_i \leq t_i \leq b_i \quad \forall i \in V_k \quad (2h)$$

$$t_i + s_i + t_{ij} \leq t_{n+i} \quad \forall i \in P_k \quad (2i)$$

$$(L_i + l_j - L_j) x_{ij} \leq 0 \quad \forall i, j \in V_k \quad (2j)$$

$$0 \leq L_i \leq L_{\max} \quad \forall i \in V_k \quad (2k)$$

$$L_0 = 0, L_{2n_k+1} = 0 \quad (2l)$$

$$x_{ij} \in \{0, 1\}, t_i \geq 0 \quad \forall i, j \in V_k \quad (2m)$$

where (2a) is the traveling cost of courier k , which consists of terms corresponding to traveling distance and traveling time. Nonnegative constants w_k^d and w_k^t are the weights that represent the importance of traveling distance and traveling time, respectively. For example, for an OC that values the time more, she can set a larger value for w_k^t . Equations (2b) and (2c) ensure that each request is served exactly once. Constraints (2d) and (2e) guarantee that all paths start at the origin and end at the destination. Equation (2f) is the flow conservation constraint. Consistency of the time variables is ensured by (2g), that is, if node j is visited immediately after node i , the arrival time at node j cannot be earlier than the arrival time at i plus the service time at i and the traveling time between i and j . The arrival time at each node is also bounded by the time window of each node, as in (2h). Precedence constraints between the

pickup and delivery nodes of each request are imposed by (2i). Consistency of load variables is ensured by (2j), where $l_j > 0$ when $j \in P_k$ and $l_j < 0$ when $j \in D_k$. Constraint (2k) imposes the load capacity, while constraint (2l) fixes the load value at the origin and destination. Finally, (2m) imposes that x_{ij} must be binary, and time variable t_i must be nonnegative.

The optimal value C_{PDP}^* of problem (2) is the optimal traveling cost for serving requests. After C_{PDP}^* has been obtained, courier k can calculate the extra cost $\bar{b}_k$ for performing the crowdsipping requests

$$\bar{b}_k = C_{PDP,k}^* - C_{OD,k} \quad (3)$$

where $C_{OD,k}$ is the traveling cost from the origin to the destination directly.

Remark 1: In general, for each OC, only a few requests will be recommended to her considering her preference and convenience.³ Therefore, the size of PDPTW solved by each OC will not be large. It can be implemented on the OC's own mobile phone through either exact solution methods or heuristics. In this way, the challenging assignment and routing problem in a large-scale crowdsipping platform can be distributed to each OC. The platform itself is able to focus on the determination of request winners and the calculation of payments.

C. Problem Formulation

In this section, mathematical models with respect to the CRA mechanism are presented in detail.

First, the problem of determining the winning bids is formulated as a winner determination problem (WDP) as follows:

$$\text{WDP: } \min \sum_{\beta_k \in S} b_k \quad (4a)$$

$$\text{s.t. } \bigcup_{\beta_k \in S} Q_k = \mathcal{N} \quad (4b)$$

which means the crowdsipping platform minimizes the total cost under the constraint that all the requests are served by the winning bids.

Problem WDP can be further reformulated into an integer program WDP-IP

$$\text{WDP-IP: } \min_{y_k} \sum_{k \in \mathcal{K}} b_k y_k \quad (5a)$$

$$\text{s.t. } \sum_{k: j \in Q_k, k \in \mathcal{K}} y_k \geq 1 \quad \forall j \in \mathcal{N} \quad (5b)$$

$$y_k \in \{0, 1\} \quad (5c)$$

where binary variable $y_k = 1$ when bid k wins, and $y_k = 0$ when k loses. The objective function (5a) represents the total cost of winning bids. Constraint (5b) ensures for every request $j \in \mathcal{N}$, there exists at least one bid that serves it.

Remark 2: Despite the reported traveling cost b_k may be different from the true cost $\bar{b}_k$, as we will show in the following sections, our proposed CRA framework can ensure truthfulness, i.e., reporting true cost is the optimal solution

³Possible recommendation methods to crowdsourced riders can refer to [19].

for each OC. Therefore, the traveling cost accounted for in the system model is in fact the true cost $\bar{b}_k$.

Next, we discuss the other key point, i.e., how to determine payments. Our goal is to design a mechanism that satisfies truthfulness and individual rationality. Before bringing in the formal definition of these two properties, we introduce the concept of dominant strategy.

Definition 2 (Dominant Strategy): A bidding strategy β_k^* of a courier k is the dominant strategy if for every alternate bidding strategy β_k and any strategy profile of other couriers β_{-k} , we have that

$$u_k(\beta_k^*, \beta_{-k}) \geq u_k(\beta_k, \beta_{-k}).$$

Then, definitions of the two desired properties follow the below definition.

Definition 3 (Truthfulness): A CRA is truthful if it is the dominant strategy for every courier $k \in \mathcal{K}$ to bid her true value $\bar{\beta}_k = (Q_k, \bar{b}_k)$. In other words, for every possible bidding strategy β_k and β_{-k} , the following inequality holds:

$$u_k(\bar{\beta}_k, \beta_{-k}) \geq u_k(\beta_k, \beta_{-k}).$$

Definition 4 (Individual Rationality): The utility of each user is nonnegative, i.e., $u_k = p_k - c_k \geq 0 \quad \forall k \in \mathcal{K}$.

Remark 3: When a mechanism is truthful, there will be no incentive for couriers to misreport their true traveling cost. When a mechanism is individual rational, participation in the mechanism can only bring benefits to the couriers, since the utility of not participating is 0.

Traditionally, in order to design a truthful combinatorial auction mechanism, the VCG mechanism can be applied [38]. However, the VCG mechanism requires problem WDP must be exactly solved to optimum, and using a nonoptimal WDP solution together with the VCG payment scheme does not yield truthful mechanisms [39]. As shown in Theorem 1, problem WDP is NP-hard, and there is no computationally efficient algorithm to solve it exactly. Consequently, the VCG mechanism is not suitable for the crowdsipping framework where a large number of couriers and requests exist. In the next section, we will pursue a non-VCG auction mechanism that is computationally efficient, truthful, and individual rational.

Theorem 1: The WDP is NP-hard.

Proof: WDP is in fact a version of the weighted set cover problem, which is well known to be NP-hard [40]. ■

IV. DESIGN OF CRA MECHANISM

In order to devise a computationally efficient mechanism, we first propose an approximate algorithm (AA) to solve the WDP. After that, a corresponding payment scheme is developed, followed by theoretical analyses of the proposed auction mechanism.

A. Approximate Algorithm for WDP

The basic idea of the AA is to greedily select the next most cost-effective bid that makes the greatest progress toward finishing all requests. Specifically, for each bid β_k , its progress

Algorithm 1 AA for WD Problem

Input: The set of pickup and delivery demands $\mathcal{N} = \{1, \dots, N\}$; the set of bids from couriers $\mathcal{B}$;
Output: The set of winning bids $\mathcal{S}$;
// Initialization
1: $\mathcal{S} \leftarrow \emptyset, \tilde{Q} \leftarrow \emptyset$;
2: **while** $\tilde{Q} \neq \mathcal{N}$ **do**
3: **for all** β_k in $\mathcal{B}$ **do**
4: **if** $Q_k \subseteq \tilde{Q}$ **then**
5: Remove β_k from $\mathcal{B}$;
6: **else**
7: $r(\beta_k) = \frac{b_k}{|Q_k - \tilde{Q}|}$;
8: **end if**
9: **end for**
10: Sort $r(\beta_k) \forall \beta_k \in \mathcal{B}$ in a nondecreasing order and denote the resulting list as $\mathcal{R}$;
11: Denote the head of $\mathcal{R}$ as $\beta_h = (Q_h, c_h)$;
12: $\mathcal{S} \leftarrow \mathcal{S} \cup \beta_h, \tilde{Q} \leftarrow \tilde{Q} \cup Q_h$;
13: Remove β_h from the set $\mathcal{B}$;
14: **end while**
15: **return** $\mathcal{S}$;

toward finishing requests is defined by the ranking criterion $r(\beta_k)$

$$r(\beta_k) = \frac{b_k}{|Q_k - \tilde{Q}|} \quad (6)$$

where $\tilde{Q}$ is the current request coverage by the winning bids. Initially, $\tilde{Q}$ is an empty set and it expands toward the whole set $\mathcal{N}$ after each iteration. $|Q_k - \tilde{Q}|$ is the number of requests that belongs to Q_k but not $\tilde{Q}$. Details of the algorithm are presented in Algorithm 1.

Algorithm 1 iterates until the current coverage $\tilde{Q}$ equals to the request set $\mathcal{N}$ (step 2). In each iteration, the bid whose preferred request set Q_k is a subset of $\tilde{Q}$ is deleted (steps 4 and 5), while the rest is ranked according to (6) (step 10). The head of the sorted set is the most cost-effective bid in this iteration, and thus is added to the winning bid set (step 12).

B. Critical Payment Scheme

After winning bids are determined based on Algorithm 1, a *critical payment* [38] scheme is introduced for calculating payments to the couriers. Different from the critical payment scheme designed for forward auction in [38], we adapt it to a reverse auction.

The detailed payment scheme is shown in Algorithm 2. In step 1, the critical payment for a winning bid β_k is set as $p_c(\beta_k) = 0$, and the request coverage set $\tilde{Q}_k$ is initialized as an empty set. After that, bid β_k is deleted from the bid set $\mathcal{B}$. Same with the process in Algorithm 1, we greedily find the winning bids with the remaining bid set $\mathcal{B}_{-k}$ (steps 2–11). In step 12, a candidate payment is defined for bid β_k

$$p^h(\beta_k) = r(\beta_k^h) \cdot |Q_k - \tilde{Q}_k^{q-1}| = b_k^h \frac{|Q_k - \tilde{Q}_k^{q-1}|}{|Q_k^h - \tilde{Q}_k^{q-1}|} \quad (7)$$

Algorithm 2 CRA Critical Payment Scheme

Input: Bid $\beta_k = (Q_k, b_k)$, other submitted bids $\mathcal{B}_{-k} = \{\beta_j | j \neq k, j \in \mathcal{K}\}$;
Output: Critical bid β_k^c , critical payment $p_c(\beta_k)$;
// Initialization
1: $p_c(\beta_k) \leftarrow 0, \tilde{Q}_k \leftarrow \emptyset$;
2: **while** $\tilde{Q}_k \neq \mathcal{N}$ **do**
3: **for all** β_j in $\mathcal{B}_{-k}$ **do**
4: **if** $Q_j \subseteq \tilde{Q}_k$ **then**
5: Remove β_j from $\mathcal{B}_{-k}$;
6: **else**
7: $r(\beta_j) = \frac{b_j}{|Q_j - \tilde{Q}_k|}$;
8: **end if**
9: **end for**
10: Sort $r(\beta_j) \forall \beta_j \in \mathcal{B}_{-k}$ in a nondecreasing order and denote the resulting list as $\mathcal{R}_k$;
11: Denote the head of $\mathcal{R}_k$ as $\beta_k^h = (Q_k^h, b_k^h)$;
12: $p^h(\beta_k) = r(\beta_k^h) \cdot |Q_k - \tilde{Q}_k^{q-1}|$;
13: **if** $p^h(\beta_k) > p_c(\beta_k)$ **then**
14: $p_c(\beta_k) = p^h(\beta_k), \beta_k^c = \beta_k^h$;
15: **end if**
16: **if** $Q_k \subseteq \tilde{Q}_k \cup Q_k^h$ **then**
17: **return** β_k^c and $p_c(\beta_k)$;
18: **end if**
19: $\tilde{Q}_k \leftarrow \tilde{Q}_k \cup Q_k^h$;
20: Remove β_k^h from $\mathcal{B}_{-k}$;
21: **end while**

where q represents the q th iteration, the winning bid $\beta_k^h = (Q_k^h, b_k^h)$ obtained in the q th iteration is the head of list $\mathcal{R}_k$, and $\tilde{Q}_k^{q-1}$ is the request coverage when the $(q-1)$ th iteration ends. If the candidate payment $p^h(\beta_k)$ at iteration q is larger than the current critical payment $p_c(\beta_k)$, the value of $p_c(\beta_k)$ is updated by $p^h(\beta_k)$, and the current critical bid is updated by β_k^h (steps 13–15). In steps 16–18, if the newly obtained winning bid β_k^h makes bid β_k “useless,” i.e., the requests served in Q_k are already covered by $\tilde{Q}_k \cup Q_k^h$, then Algorithm 2 ends, returns the critical bid β_k^c and the critical payment $p_c(\beta_k)$. If Algorithm 2 does not end in step 17, the current coverage set is updated (step 19), and the head bid β_k^h is deleted from $\mathcal{B}_{-k}$ (step 20).

To summarize, the critical payment of bid β_k is the largest candidate payment $p^h(\beta_k)$ until bid β_k is no longer “useless.” Finally, the payments to couriers are defined as follows:

$$p_k = \begin{cases} p_c(\beta_k), & \beta_k \in \mathcal{S} \\ 0, & \beta_k \notin \mathcal{S} \end{cases} \quad (8)$$

where winning couriers are paid their critical payments, and losing couriers receive no payment.

C. Theoretical Analyses

With the winner determination scheme and courier payment scheme at hand, we present rigorous theoretical analyses to prove our proposed auction mechanism is truthful, individual rational, and computationally efficient.

First, the definition of monotonicity is introduced to facilitate the analysis, and a lemma follows to show the monotonicity of Algorithm 1.

Definition 5 (Monotonicity): For each bid $\beta_k = (Q_k, b_k)$, if β_k wins, then bid $\hat{\beta}_k = (Q_k, \hat{b}_k)$ also wins, where $\hat{b}_k \leq b_k$.

Lemma 1: The approximate winner determination algorithm satisfies monotonicity.

Proof: Refer to Appendix A. ■

Next, Lemma 2 introduces the definition of *critical value* and shows its existence in the CRA mechanism.

Lemma 2: For a bidder k , there exists a *critical value* $c(\beta_k)$ such that if bidder k bids a cost value b_k which is lower than or equal to $c(\beta_k)$, she will win; otherwise, she will lose.

Proof: Refer to Appendix B. ■

According to Lemma 3, the critical payment is exactly a critical value.

Lemma 3: The critical payment $p_c(\beta_k)$ to winning bidder k in Algorithm 2 is the critical value for this bidder.

Proof: Refer to Appendix C. ■

Lemma 4: Under the critical payment scheme, the utility of a *truthful* bidder is always nonnegative.

Proof: Refer to Appendix D. ■

By Lemmas 2–4, the truthfulness of the CRA mechanism can be proved.

Theorem 2: The proposed CRA mechanism is truthful.

Proof: Refer to Appendix E. ■

Moreover, the individual rationality of the CRA mechanism also follows the below corollary.

Corollary 1: The proposed CRA mechanism satisfies individual rationality.

Proof: This corollary follows directly from Theorem 2, Lemma 4, and Definition 4. ■

Furthermore, Theorem 3 provides the approximation ratio of Algorithm 1.

Theorem 3: Algorithm 1 is an $O(\mathcal{H}(m))$ -approximation algorithm, where $m = \max_{k \in \mathcal{K}} |Q_k|$ is the largest size of bids, and $\mathcal{H}(m)$ is the m th Harmonic number, i.e., $\mathcal{H}(m) = \sum_{i=1}^m \frac{1}{i} \approx \log m + \gamma$, and γ is Euler's constant.

Proof: Refer to Appendix F. ■

Finally, Theorem 4 shows the computation efficiency of our proposed mechanism.

Theorem 4: The proposed CRA mechanism has polynomial-time computation complexity.

Proof: Refer to Appendix G. ■

V. NUMERICAL RESULTS

In this section, the performance of the CRA mechanism is evaluated through extensive simulations. All experiments are implemented using MATLAB R2021a on a desktop with 2.5-GHz Intel Core i7 CPU and 16-GB Memory.

A. Solomon Test Case-Based Simulation

The widely used Solomon benchmark problems [41] are first selected as our test instances. Taking the “R101” problem set with 25 customers as an example,⁴ since Solomon test cases

⁴Note that using other problem sets achieves similar results.

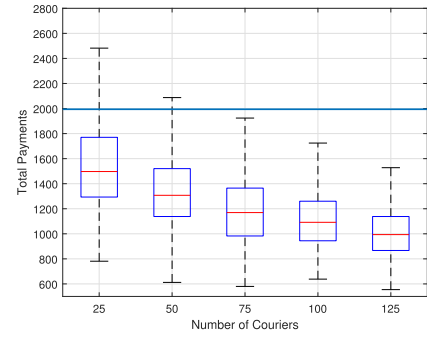


Fig. 5. Total payments to the couriers when the number of couriers increases.

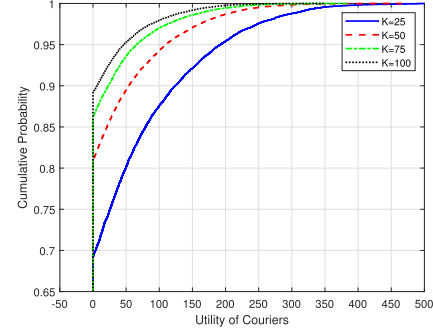


Fig. 6. Cumulative distribution of couriers' utility (Solomon test case).

are for the vehicle routing problem, we adapt it to the PDPTW problem based on the method in [42]. In this way, the R101 problem set is transformed into a pickup and delivery test case with 13 requests. The origins and destinations of couriers are randomly generated with uniform distribution over the service area of pickup and delivery requests. The bid cost of each courier is calculated by solving the PDPTW problem (2). The capacity of each courier is set as 30.

Fig. 5 depicts the total payments paid by the platform when the number of couriers increases from 25 to 125. For each courier number, 500 random cases are simulated and the corresponding box plot is presented. From this figure, we can observe that the average total payment decreases with the number of couriers. The reason is when more OCs are available, the platform has better chance to find more cost-effective couriers that serve requests along their route to destination. In Fig. 5, there also exists a horizontal straight line around 2000. This represents the traveling cost when the platform employs self-owned vehicles to serve requests. The depot of self-owned vehicles is assumed to be the center of the service area, the number of available vehicles is set as 30 and their capacities are assumed to be 50. Therefore, we can conclude that employing OCs is cheaper than employing self-owned vehicles in almost all cases.

Fig. 6 plots the cumulative distribution of couriers' utility. It is easy to observe that all the couriers acquire nonnegative utility under the proposed mechanism, thus the property of individual rationality is verified.

Fig. 7 illustrates how the courier's utility changes when her bid cost varies. The horizontal coordinate represents the percentage of cost misreporting, where 0% means the courier

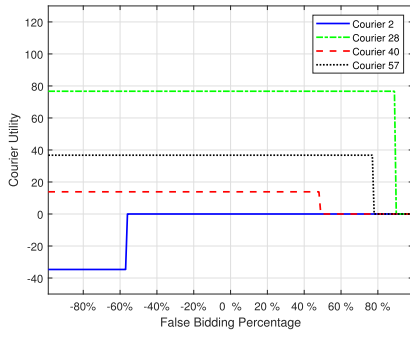


Fig. 7. Courier utility when bid cost varies (Solomon test case).

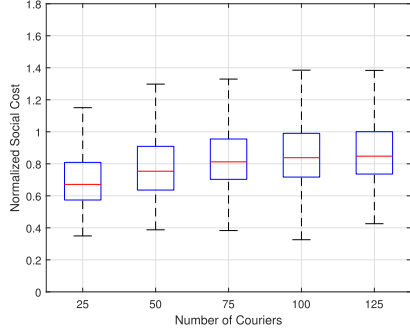


Fig. 8. Normalized social cost of CRA mechanism. The social cost of Vickery auction mechanism is set as 1.

reports her bid cost truthfully, 20% means she claims her cost to be 120% of the true cost, and -20% means her claimed cost is 80% of the true cost. In this figure, we can see that reporting falsely cannot improve the courier's utility.⁵ Hence, our CRA mechanism satisfies the property of truthfulness.

After that, we adapt the multiattribute reverse Vickery auction mechanism studied in [35] to this test case, and compare with our proposed CRA mechanism. Similar to the settings in Fig. 5, 500 random cases are simulated, and the corresponding box plot is presented. By setting the social cost of the Vickery auction mechanism as 1, Fig. 8 illustrates the normalized social cost using the CRA mechanism. For the majority of cases, the CRA mechanism achieves less social cost than the Vickery mechanism. The reason is that using a combinatorial auction and allowing package bidding can take better advantage of economies of scope. Similarly, in Fig. 9, we can observe that for most cases, the total payment of the crowdshipping platform to the couriers when using CRA is also less than that by using Vickery auction, leading to a substantial financial benefit for the platform.

B. Random Test Instance

To further validate our proposed CRA mechanism, a larger-scale random test instance is studied. Similar to the settings in [14], a square service region is used, where origins and destinations of OCs, as well as pickup and delivery nodes of requests, are all assumed to be uniformly distributed over this region. The size of the square region is $60 \times 60 \text{ km}^2$.

⁵Here, we only select several couriers as an example. When other couriers are selected, this result still holds.

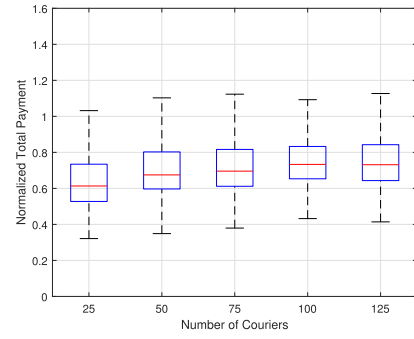


Fig. 9. Normalized total payment of the crowdshipping platform using the CRA mechanism. The payment with Vickery auction mechanism is set as 1.

TABLE II
AVERAGE RUNTIME

$N = K$	AA runtime [sec]	OA runtime [sec]
450	0.3844	51.2642
500	0.4965	100.0043
550	0.5863	110.1633
600	0.7458	483.5649
650	0.9104	2433.8
700	1.0735	5065.5

We first compare the computational properties between the proposed CRA mechanism and the widely used VCG mechanism. The WDP solution algorithm of CRA is denoted as the AA, i.e., Algorithm 1, while an optimal algorithm (OA) is required to solve the WDP in the VCG mechanism. Analyzing Algorithm 2, we can see that to calculate the payment of each winning bid, one slightly modified process of AA is also performed, where the difference from AA in Algorithm 1 is that here the winning bid itself is excluded. Similarly, calculating the payment of each winning bid in the VCG mechanism also needs to solve a WDP without the winning bid itself. Therefore, we can compare the computational property of CRA and VCG by comparing AA and OA instead. The number of requests N and couriers K are increased from 450 to 700, where $N = K$. For each value of N , both algorithms run for ten times. Table II provides the average runtime of AA and OA. When the size of the test instance grows to 650 or 700, the computation time of OA becomes prohibitively long. Thus, OA is no longer applicable for crowdshipping with hundreds of couriers. As for the optimal value achieved by both algorithms, Fig. 10 demonstrates that the approximation ratio is around 1.1 to 1.2. Here, the approximation ratio is defined as the cost achieved by AA divided by the optimal cost. In summary, when the crowdshipping system becomes large, our proposed CRA mechanism achieves near-optimal cost with significantly shorter time than the VCG mechanism.

Next, we fix the number of requests N and vary the number of couriers K . In Fig. 11, N is set as 500. When K increases, the total payments from the platform decrease, since more cost-effective couriers are available.

In Fig. 12, the distribution of couriers' utility is illustrated. We can observe that all the couriers can obtain nonnegative utility, verifying the satisfaction of individual rationality.

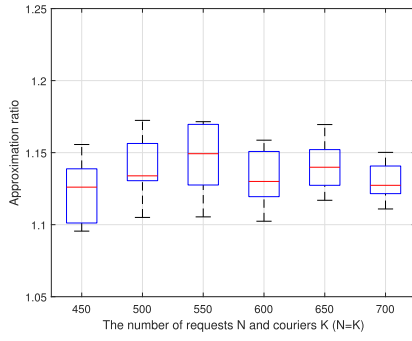


Fig. 10. Approximation ratio of Algorithm 1 over the OA.

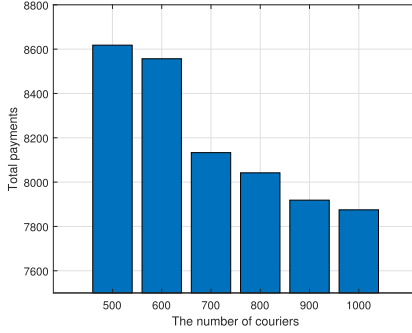


Fig. 11. Average total payments when the number of couriers varies while the number of requests fixes.

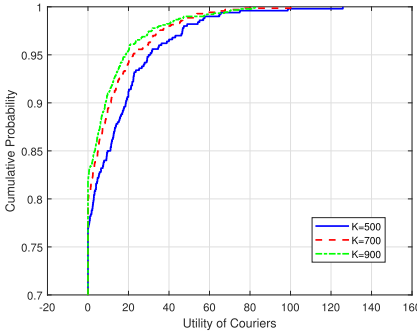


Fig. 12. Cumulative distribution of couriers' utility (random test case).

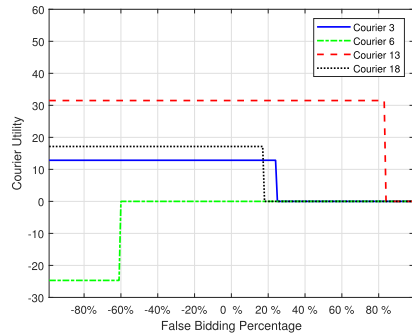


Fig. 13. Courier utility when bid cost varies (random test case).

Finally, Fig. 13 shows the utility of several couriers when they claim different costs. No matter what cost value they claim, they cannot achieve a larger utility than reporting truthfully. Hence, the property of truthfulness also holds.

VI. CONCLUSION

This article has investigated the problem that a crowdshipping platform recruits OCs to serve the parcel pickup and delivery requests. We introduced a combinatorial reverse auction framework to model the interactions between the platform and couriers. An auction mechanism that achieves near-optimal social cost was designed. Meanwhile, truthfulness, individual rationality, polynomial computation time, and the approximation ratio of this mechanism were rigorously proved. After extensive simulations, the desired properties of our proposed auction mechanism have been validated.

In the future, we plan to further extend this work in several directions. First, we intend to implement the proposed mechanism in real applications. Second, considering uncertainty in traveling time can be an interesting problem to investigate. Third, we aim to study coalition-proof auction mechanisms to prevent couriers from collusion.

APPENDIX A PROOF OF LEMMA 1

Suppose that bid β_k has won in the q th iteration. A number of winning bids have already been determined in the previous iterations. We can store the winning bids in a sorted list $L = \{\beta^1, \beta^2, \dots\}$ with the order of determination, i.e., β^1 is the first determined winning bids and β^2 is the second one. Then, β_k is the q th element in list L . Replacing β_k with another bid $\hat{\beta}_k = (Q_k, \hat{b}_k)$, where $\hat{b}_k \leq b_k$, two cases are discussed.

- 1) If β_k is determined before the q th element in L , since the q th element of L still wins, then $\hat{\beta}_k$ wins trivially.
- 2) If $\hat{\beta}_k$ is not determined before the q th element in L , then by the ranking condition (6) and Algorithm 1, $\hat{\beta}_k$ has higher priority to be determined than β_k , and it will win in the q th iteration.

Monotonicity of Algorithm 1 follows from Definition 5.

APPENDIX B PROOF OF LEMMA 2

By monotonicity, there exists a value c_k such that when bidder k bids in the set $[0, c_k]$ (c_k may equal to 0), she always wins. Let $c(\beta_k)$ be the largest value of such c_k , then when she bids b_k which is lower than or equal to $c(\beta_k)$, she will win. Now suppose there is another bid value $\hat{c}_k > c(\beta_k)$ that wins the auction, according to monotonicity, for all values less than $\hat{c}_k$, bidder k still wins the auction, which violates the largest value assumption of $c(\beta_k)$. Hence, we can claim that when bidder k bids a cost value larger than $c(\beta_k)$, she will lose.

APPENDIX C PROOF OF LEMMA 3

Assume that the critical bid β_k^c wins at the q_c th iteration, and Algorithm 2 ends at the q_{end} th iteration, i.e., at the q_{end} th iteration, condition $Q_k \subseteq \tilde{Q}_k \cup Q_k^h$ holds. The critical payment $p_c(\beta_k)$ to winning bidder k is equal to $r(\beta_k^c) \cdot |Q_k - \tilde{Q}_k^{q_c-1}|$, where $r(\beta_k^c)$ is the ranking value of critical bid β_k^c .

On the one hand, assume bidder k bids a value $b_k \leq p_c(\beta_k)$, according to (6), its ranking value $r(Q_k, \hat{b}_k)$ at the q_c th

iteration is

$$r(Q_k, \hat{b}_k) \leq \frac{r(\beta_k^c) \cdot |Q_k - \tilde{Q}_k^{q_c-1}|}{|Q_k - \tilde{Q}_k^{q_c-1}|} = r(\beta_k^c). \quad (9)$$

Therefore, the ranking priority of this bid is higher than the critical bid. As the critical bid wins at the q_c th iteration when bid (Q_k, b_k) is absent, bidding $(Q_k, \hat{b}_k)$ will clearly win the auction too.

On the other hand, assume bidder k bids a value $\tilde{b}_k > p_c(\beta_k)$. Since $p_c(\beta_k)$ is the largest value among all the $p^h(\beta_k)$ until the q_{end} th iteration, according to (6), we have

$$\begin{aligned} \frac{\tilde{b}_k}{|Q_k - \tilde{Q}_k^{q-1}|} &> \frac{p_c(\beta_k)}{|Q_k - \tilde{Q}_k^{q-1}|} \\ &\geq \frac{p^h(\beta_k)}{|Q_k - \tilde{Q}_k^{q-1}|} = r(\beta_k^h) \quad \forall q \leq q_{\text{end}} \end{aligned} \quad (10)$$

where β_k^h is the head of $\mathcal{R}_k$ in the q th iteration. Hence, the ranking priority of bid $(Q_k, \tilde{b}_k)$ is lower than all the bids before the $(q_{\text{end}} + 1)$ th iteration. That is, it will not win the auction before the $(q_{\text{end}} + 1)$ th iteration. Moreover, after the q_{end} th iteration, Q_k is no longer useful and thus will not win the following iterations.

Based on the definition of critical value in Lemma 2, the critical payment is also the critical value for bidder k .

APPENDIX D PROOF OF LEMMA 4

Under the critical payment scheme, both the delivery cost and the received payment of a losing bidder are zero, then her utility is also zero. As for a winning bidder k , since she is truthful, her bid will be $\beta_k = (Q_k, \bar{b}_k)$. By Lemmas 2 and 3, when bidder k wins, her bid cost must be less than or equal to the critical payment $p_c(\beta_k)$, i.e., $\bar{b}_k \leq p_c(\beta_k)$. Hence, her utility is $u_k = p_c(\beta_k) - \bar{b}_k \geq 0$.

APPENDIX E PROOF OF THEOREM 2

Denote the utility of bidder k as $\bar{u}_k$ when she bids truthfully with $\bar{\beta}_k = (Q_k, \bar{b}_k)$, and the utility as u_k when she bids a false value $\beta_k = (Q_k, b_k)$, $b_k \neq \bar{b}_k$. Four different cases are considered.

- 1) *Bid $\bar{\beta}_k$ Wins, and β_k Loses:* By Lemma 4, $\bar{u}_k \geq 0$, and $u_k = 0$, then we have $\bar{u}_k \geq u_k$.
- 2) *Bid $\bar{\beta}_k$ Loses, and β_k Wins:* By Lemma 3, this case can only happen if $\bar{b}_k > p_c(\beta_k) \geq b_k$. Since $\bar{\beta}_k$ loses the auction, $\bar{u}_k = 0$. Despite bidder k can bid a false value b_k , her true cost value is still $\bar{b}_k$, then we have $u_k = p_c(\beta_k) - \bar{b}_k < 0$. Hence, $\bar{u}_k > u_k$.
- 3) *Both $\bar{\beta}_k$ and β_k Win:* Based on the critical payment scheme, the two bids receive the same payment and have the same cost value, therefore, $\bar{u}_k = u_k$.
- 4) *Both $\bar{\beta}_k$ and β_k Lose:* $\bar{u}_k = u_k = 0$ in this case.

To summarize, $\bar{u}_k \geq u_k$ always holds, thus bidding truthfully is a dominant strategy.

APPENDIX F PROOF OF THEOREM 3

We rewrite (5b) in the WD problem (5) as follows:

$$\sum_{j=1}^N a_{jk} y_k \geq 1 \quad \forall k \in \mathcal{K} \quad (11)$$

where $a_{jk} = 1$ when pickup and delivery demand $j \in Q_k$ and $a_{jk} = 0$ otherwise.

In each iteration of Algorithm 1, one bid is selected as the winner. Without loss of generality, let the bids selected in the first i iterations be $\{1, 2, \dots, i\}$, which is a subset of the bidder set $\mathcal{K}$. After each iteration, some uncovered pickup and delivery demands in $\mathcal{N}$ are covered by the new winner. With the coverage of winning bids $\tilde{Q}$ becoming wider, the number of useful elements⁶ in each remaining bid may decrease. Let Q_k^i be the set of remaining useful elements of bidder k at the beginning of iteration i , i.e., $Q_k^i \subseteq Q_k \setminus \tilde{Q}$. Denote the size of Q_k^i as $s_k^i = |Q_k^i|$. When bidder i is selected at the i th iteration, the cost it adds to the total cost is b_i . The average cost of each useful element is defined as b_i/s_i^i . Due to the greedy determination scheme of Algorithm 1, at iteration i , the average element cost of bidder i must be the lowest among all bids

$$\frac{b_i}{s_i^i} \leq \frac{b_k}{s_k^i} \quad \forall k \in \mathcal{K}. \quad (12)$$

Assume that there are I iterations altogether. Each pickup and delivery demand $j \in \mathcal{N}$ belongs to one set Q_i^i , $i = 1, 2, \dots, I$, so for $j \in Q_i^i$, the cost that demand j adds to the total cost is defined as z_j , which satisfies $z_j = b_i/s_i^i$, $j \in Q_i^i$. Then, the total cost of WD problem (5) using Algorithm 1 is

$$C_{\text{appr}} = \sum_{j=1}^N z_j. \quad (13)$$

Observing that $Q_k \cap Q_i^i = Q_k^i \setminus Q_k^{i+1}$, we have

$$\begin{aligned} \sum_{j=1}^N a_{jk} z_j &= \sum_{i=1}^I \sum_{j \in Q_k \cap Q_i^i} z_j = \sum_{i=1}^I \sum_{j \in Q_k^i \setminus Q_k^{i+1}} z_j \\ &= \sum_{i=1}^I (s_k^i - s_k^{i+1}) \frac{b_i}{s_i^i}. \end{aligned} \quad (14)$$

For a given bid Q_k , let p be the largest index such that $s_k^p > 0$, then

$$\begin{aligned} \sum_{j=1}^N a_{jk} z_j &= \sum_{i=1}^p (s_k^i - s_k^{i+1}) \frac{b_i}{s_i^i} \leq b_k \sum_{i=1}^p \frac{s_k^i - s_k^{i+1}}{s_i^i} \\ &\leq b_k \sum_{i=1}^p (\mathcal{H}(s_k^i) - \mathcal{H}(s_k^{i+1})) \leq b_k \mathcal{H}(s_k^1) \leq b_k \mathcal{H}(m) \end{aligned} \quad (15)$$

⁶Useful elements refer to the elements of bids that still can be selected to cover the unsatisfied demands. When all the elements of a bid Q_k have already been covered by other bids, then all the elements of Q_k are useless.

where the first inequality follows from (12), and the second inequality follows from the fact that:

$$\mathcal{H}(s_k^i) - \mathcal{H}(s_k^{i+1}) = \sum_{s_k^{i+1}+1}^{s_k^i} \frac{1}{i} \geq \sum_{s_k^{i+1}+1}^{s_k^i} \frac{1}{s_k^i} = \frac{s_k^i - s_k^{i+1}}{s_k^i}. \quad (16)$$

After that, we can establish that for the optimal solution y_k^* of WD problem (5), the following result holds:

$$\begin{aligned} \sum_{k=1}^K \mathcal{H}(m) b_k y_k^* &\geq \sum_{k=1}^K \left(\sum_{j=1}^N a_{jk} z_j \right) y_k^* \\ &= \sum_{j=1}^N \left(\sum_{k=1}^K a_{jk} y_k^* \right) z_j \geq \sum_{j=1}^N z_j = C_{\text{appr}} \end{aligned} \quad (17)$$

where the first inequality follows from (15), and the last inequality comes from (11). Since the optimal value of the WD problem is $C_{\text{opt}} = \sum_{k=1}^K b_k y_k^*$, we have

$$C_{\text{appr}} \leq \mathcal{H}(m) C_{\text{opt}}. \quad (18)$$

I.e., Algorithm 1 achieves an approximation ratio of $\mathcal{H}(m)$.

APPENDIX G PROOF OF THEOREM 4

For Algorithm 1, the time complexity of the *for* loop is $O(K)$, where K is the number of bidders. The operation of sorting in line 10 has complexity $O(K \log K)$. Since in each iteration, at least one demand in $\mathcal{N}$ is covered, the *while* loop will run for at most N times. Hence, the total time complexity of Algorithm 1 is $O(NK \log K)$. For Algorithm 2, we can easily obtain similar results. Therefore, the polynomial-time complexity of the proposed CRA mechanism is proved.

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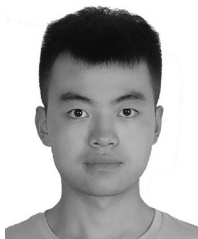
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